

Today

ODEs

Euler's

Announcements
Good Luck on Exams!
Pdf + Code

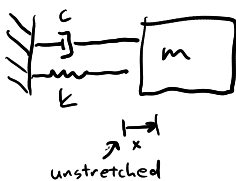
$$\dot{x} \equiv \frac{dx}{dt} = ax$$

$$\frac{dx}{dt} - ax = 0$$

$$x = C e^{at}$$

determined by boundary condition

$$x = x_0 e^{at}$$



char. eqn.

$$f = ma$$

$$-kx - c\dot{x} = m\ddot{x}$$

$$m\ddot{x} + c\dot{x} + kx = 0$$

$$m\lambda^2 + c\lambda + k = 0$$

$$\lambda = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m}$$

$$\ddot{x} = \frac{d^2x}{dt^2}$$

differential
ordinary
linear
second order
homogeneous

equation

differential: involves derivatives

ordinary: only 1 indep. variable (t) as opposed to PDE

linear: each term is linear (no x^2 , $\sin x$)

first order: only involves first derivatives

homogeneous: no "forcing" / 0 rhs

Damping

Over: λ : 2 distinct real

$$x = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t}$$

Critically $\lambda = -\frac{c}{2m}$ (repeated)

$$x = (C_1 + C_2 t) e^{\lambda t}$$

Under $\lambda = \sigma \pm i\omega$

$$x = e^{\sigma t} (C_1 \sin(\omega t) + C_2 \cos(\omega t))$$

Roughly: solutions to linear ODEs are exponentials or growing or decaying oscillations (sin/cos)

Nonlinear ODEs ???

Numerical Integration

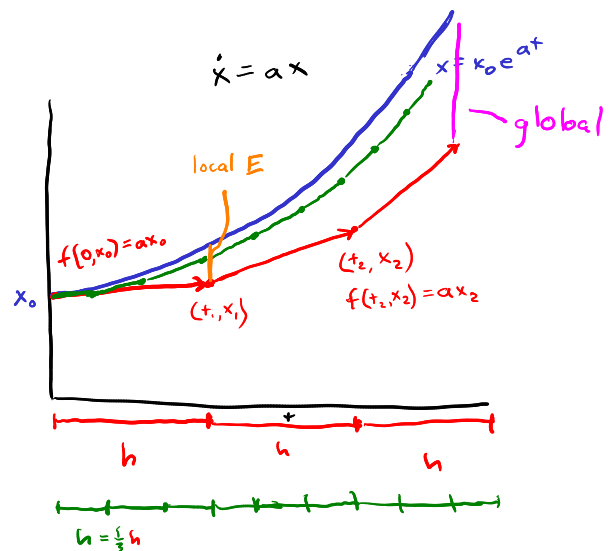
$$\dot{x} = f(t, x)$$

dynamics

Euler's Method

$$t_{i+1} = t_i + h \quad \leftarrow \text{step size}$$

$$x_{i+1} = x_i + f(t_i, x_i) h$$



Error Analysis

Truncation (Taylor Series)

Roundoff (floating point
sig figs)

local (each step)
global (after all steps)

Taylor Series

$$x_{i+1} = x_i + f(t_i, x_i)h + \frac{\frac{\partial f}{\partial t}(t_i, x_i)}{2!} h^2 + \frac{\frac{\partial^2 f}{\partial t^2}(t_i, x_i)}{3!} h^3 \dots$$

Euler update

$$\text{Local } E_f = \frac{\frac{\partial f}{\partial t}(t_i, x_i)}{2!} h^2 + O(h^3)$$

$$\text{Local } E_f = O(h^2) \text{ as } h \rightarrow 0$$

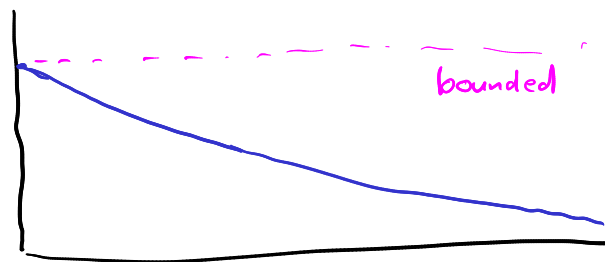
$$\text{Global } E_f = O(h) \text{ as } h \rightarrow 0$$

Stability of numerical ODE integrators

A numerical integrator is said to be unstable if errors grow exponentially for a problem with a bounded solution.

$$\dot{x} = -ax \quad a > 0$$

$$x = x_0 e^{-at}$$



Euler's $f(t, x_i)$

$$x_{i+1} = x_i - ax_i h \\ = x_i (1 - ah)$$

$$x_n = x_0 \underbrace{(1 - ah)^n}_{\text{amplification factor}}$$

x_n shrinks if $|1 - ah| < 1$

grows exponentially if $|1 - ah| > 1$

Stability depends on 3 factors: ODE
Integration method
step size h

For $\dot{x} = -ax \quad a > 0$

$\begin{cases} h > \frac{2}{a} & \text{Euler's unstable} \\ h < \frac{2}{a} & \text{Euler's stable} \end{cases}$

Euler's method is conditionally stable

What about systems? higher order?

$$\rightarrow \boxed{\dot{\vec{x}} = f(t, \vec{x})}$$

$$m\ddot{x} + c\dot{x} + kx = 0$$

$$v \equiv \dot{x}$$

$$\dot{v} = -\frac{c}{m}v - \frac{k}{m}x$$

$$\vec{x} = \begin{bmatrix} x \\ v \end{bmatrix}$$

↑
state vector

$$f(t, \vec{x}) = \begin{bmatrix} \dot{x} \\ \dot{v} \end{bmatrix} = \begin{bmatrix} v \\ -\frac{c}{m}v - \frac{k}{m}x \end{bmatrix}$$

↑
state derivative